

9th-semester Systems Engineering student at USM, Caracas. I have designed and deployed 2 systems in active production with real users (multi-store ERP and AI academic repository) and developed complex systems like an automated academic manager (EduFlow) and an AI heuristic phishing detector. Master of the full development lifecycle using React, Django, Node.js, Python, and relational databases.

// PROJECTS

Featured Projects

Vesta Retail ERP

[↗ Demo](#)[GitHub](#)

▸ Active production ERP/POS system for multi-store inventory management

Multi-store enterprise platform with distributed SPA architecture. Real-time inventory synchronization and concurrent invoicing using ultra-low latency WebSockets (Socket.io) and a relational MySQL database with Sequelize ORM optimized for robust ACID transactions.

React 19

Node.js

Socket.io

MySQL

Sequelize

Express

Academic Repository + AI

[↗ Demo](#)[GitHub](#)

▸ School system in production – automates academic research preservation

Premium digital system for Ntra. Sra. de Fátima School. Automates preservation and intelligent analysis of scientific projects with RBAC auth (Director/Teacher), immutable auditing module, automatic Google Drive synchronization, and PDF analysis via Gemini 2.0 Flash.

React 19

Node.js

Gemini 2.0

MySQL

JWT / RBAC

Drive API

EduFlow

[↗ Demo](#)[GitHub](#)

• Academic management and automated invoicing platform for diplomas and courses

Enterprise-grade web platform for courses administration. Automates enrollment management, real-time class attendance control, payments, and automatic PDF invoicing sent via email using background tasks (Celery & Redis) under a secure Role-Based Access Control (RBAC) scheme.

[Python/Django](#)[PostgreSQL](#)[Celery / Redis](#)[Tailwind CSS](#)[Docker](#)

PhishShield AI

[↗ Demo](#)[GitHub](#)

• Metrics: 96.7% accuracy in heuristic threat detection and false positive rate reduced to 1.2%

Defensive cybersecurity platform (Software Engineering course project, USM). Integrates a custom Explainable AI heuristic classification engine, concurrent reputation checks (VirusTotal/PhishTank APIs), a high-performance concurrent multi-threaded scraper using ThreadPoolExecutor, and SQLite optimized in 4NF.

[Python 3.10+](#)[Flask](#)[SQLite 4NF](#)[VirusTotal API](#)[BeautifulSoup4](#)[Cybersecurity](#)

// STACK

Technical Skills

FRONTEND

React 19 · Vite · JavaScript ES6+ · HTML5 · CSS3 /
CSS Modules · Tailwind CSS · Framer Motion ·
Three.js

BACKEND

Node.js · Express · Python · Flask · Django · REST
APIs · WebSockets (Socket.io) · JWT

DATABASES

MySQL · PostgreSQL · SQLite (up to 4NF) ·
Sequelize · Advanced relational modeling

AI / OTHERS

Google Gemini 2.0 SDK · OpenCV · Pandas ·
BeautifulSoup4 · VirusTotal API · Google Drive API ·
Git · GitHub

Professional Experience

Jun 2026 – Jul 2026 **Backend Developer / Creator**

EduFlow · Learning & Research Project

Designed and developed a robust academic administration web platform using Django and PostgreSQL. Implemented task queues with Celery/Redis for automated email invoicing and Docker containerization.

Apr 2026 – present **Software Developer**

Nuestra Señora de Fátima School · Community Service

Design and implementation of a local network (LAN) Full-Stack web system for research document management. Automates academic workflows with secure JWT authentication, role-based access control, and daily backups automatically synced to Google Drive.

Dec 2025 – Apr 2026 **Freelance Full-Stack Developer**

Vesta Retail ERP · Independent Project

Designed, developed, and deployed to production a multi-store ERP platform that reduced estimated physical stock reconciliation time by ~35%. Automated invoicing workflows and generated daily cash closing reports synchronized in real-time between branches.

Oct 2025 – Nov 2025 **Cybersecurity Software Developer (Academic)**

PhishShield AI · USM Software Engineering Project

Developed an intelligent URL phishing threat analyzer. Designed and implemented a heuristic classification engine powered by Explainable AI, concurrent global reputation API queries, and a SQLite database normalized in 4NF, achieving 96.7% accuracy in tests.

Courses & Certifications

UNEWEB, 2025 **Programming Diploma — Python (Flask/Django) · JavaScript · AI for Programmers · MySQL · PostgreSQL**

UCV, 2023 **C++ Programming — Levels 1 & 2**

CEBA UCV, 2025 **Basic English**

Benllisoftware, 2021 **Photoshop CC Level 1 — Graphic Design**

Education

2022 – present **Systems Engineering**

Santa Maria University (USM) · Caracas · 9th Semester

Studies in Full-Stack web development, relational database systems, software engineering, cybersecurity, and computer vision.